

7 The Renaissance in Europe

7.1 Transmission of Knowledge and Notational Development

Between the fall of Rome in 476 and the early renaissance⁴⁵ (c. 1100) came Europe's dark ages. The simplistic view is that this was a time of limited learning and technological progress, though reality was more complex. Learning and scholarship persisted in monasteries, though new research took second place to a conservative focus on preserving the wisdom of the ancients.

By 1100, some smaller kingdoms were starting to come together in more stable arrangements. The maps below show the consolidation of the major feudal societies⁴⁶ of medieval Europe: France, England, the Holy Roman Empire, Poland, Austria, etc. While it would have meant little to most peasants, a large stable nation can support a significant elite population with the time and money to pursue education, and fund libraries and universities. By the 1700s, the borders of western Europe are largely recognizable; political and social organization had expanded so that a much larger proportion of the population—though small in comparison to today—could take advantage of and contribute to the growth of knowledge. The European renaissance is often contrasted with a decline in Islamic power, but again the story is more complex: Islam retreats from Spain while the Ottoman Empire becomes dominant in south-east Europe.



Between 900 and 1300, the population of Europe roughly tripled to around 100 million. Trade increased, along with which came knowledge.⁴⁷ Learning (and land) came via wars with Islam (the Crusades, Spain, etc.). It helped that Islamic scholars so venerated the Greeks; Europeans could tell themselves that they were merely 'reclaiming' ancient knowledge which had been 'stolen' by their cultural and religious enemies. The fall of Constantinople to Mehmet in 1453 marks both the high point of European Islamic conquest and the start of the decline of Islamic scientific dominance. Many intellectuals fled Constantinople—where the Byzantines preserved much of Alexandria's learning and were more exposed to eastern ideas—for Rome, helping to further fuel developments. With powerful enemies to the east, Europeans began travelling greater distances by sea,⁴⁸ beginning the colonial era of global European empire.

⁴⁵Literally *rebirth*. Dates vary by location and discipline (Italy vs. France, art vs. philosophy, etc.) but a wide net would encompass the 12th to 17th centuries.

⁴⁶An arrangement where powerful landowners could demand service (rent, food, warriors) from their tenants.

⁴⁷Venice was a particularly important trading hub. From here, Marco Polo (1254–1324), perhaps the most famous trader of the period, travelled the silk road to China.

⁴⁸Christopher Columbus (born Genoa 1451) 'discovered' America in 1492 while looking for sea routes to Asia.

Education in the Renaissance

Scientific and philosophical progress was spurred by the translation of works from Arabic and Greek into Latin, with the first universities being formed to teach this canon: Bologna 1088, Paris 1150, and Oxford 1167. The typical student was a young man of wealth who had been privately tutored in grammar, logic & rhetoric (the *trivium*). At university he would study the Greek-influenced *quadrivium* (geometry, astronomy, arithmetic & music). While Islamic improvements were incorporated, scholars gave pre-eminent credit to the Greeks: Euclid for geometry, Aristotle for logic/physics, Hippocrates/Galen for medicine, Ptolemy for astronomy. Early universities were often funded by the Church and 'research,' was more likely to involve the justification of biblical passages using Aristotle than the conduct of experiments.

Leonardo da Pisa (Fibonacci c. 1175–1250)

Fibonacci⁴⁹ likely first encountered the Hindu–Arabic numerals while trading with his father in North Africa. He was impressed by the ease of calculation they afforded and is the first European known to use them (contemporary Europeans used Roman numerals and Egyptian fractions). Fibonacci's 1202 text *Liber Abaci* was written to instruct traders in their use. The first page below explains how to compute with decimal fractions, with the two columns at the bottom of the page showing how to repeatedly multiply 100 (and then 10) by the fraction $\frac{9}{10}$. Thus:

100, 90, 81, $\frac{9}{10}72 (= 72.9)$, $\frac{1}{10}\frac{6}{10}65 (= 65.61)$, $\frac{9}{10}\frac{4}{10}\frac{0}{10}59 (= 59.049)$, etc.



⁴⁹The name was given to Leonardo by French scholars of the 1800s: *filius Bonacci* means son of Bonacci.

Note how the fractional part is written in reverse (relative to modern practice) on the left, using a bar to separate numerator and denominator: the Indians wrote fractions without a bar and it is thought Islamic scholars inserted it for clarity in the 1100s.

The second picture is of Fibonacci's famous sequence, which he illustrates using the famous problem of breeding rabbits. Read top-to-bottom: 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377. While named for him, the sequence had been discussed for at least 1000 in India. Amongst other inheritances from the Hindu-Arabic tradition, Fibonacci is the first known European to work with negative numbers, provided these represented deficiencies or debts in accounting.

Algebraic Notation and Development

To a modern reader, the most obvious mathematical development of the renaissance is notational. Hindu-Arabic numerals and fractional notation were cutting-edge for the 1200s, but essentially everything else in Fibonacci's work was written in sentences. Over the next 500 years, notational improvements developed which eventually allowed algebra to eclipse geometry as the primary language of reasoning. Here is a very brief summary.

Italian Abacists, 14th C. This group continued Fibonacci's advocacy for the Hindu-Arabic system against the traditional use of Roman numerals, and also for the use of accompanying algorithms. Their approach was highly practical and applied largely for the conduct of trade. Here is a typical problem described by the group:

The *lira* earns three *denarii* a month in interest. How much will sixty *lire* earn in eight months?

The Abacists introduced various shorthands for unknowns and some mathematical operations: *cosa* ('thing') was used for an unknown; *censo*, *cubo* and *radice* meant, respectively, square, cube and (square-)root. These expressions could be combined, for example 'ce cu' (read *censo di cubo*) referred to the sixth power of an unknown $(x^3)^2 = x^6$.

Luca Pacioli, Italy late 1400s. Introduced $\bar{p}$, $\bar{m}$ (*piu*, *meno*) for plus and minus. For example $8\bar{m}2$ denoted eight minus two.

Nicolas Chuquet, France 1484. His text *Triparty en la science des nombres* borrowed Pacioli's $\bar{p}$ and $\bar{m}$, and introduced an *R*-notation for roots. For instance R^47 meant $\sqrt[4]{7}$, while

$\sqrt[5]{4 - \sqrt{2}}$ would be written $R^5\underline{4\bar{m}R2}$

Chuquet underline terns to indicate grouping in the way we now use parentheses.

Christoff Rudolff, Vienna 1520s. Introduced symbols similar to x and ζ for an unknown and its square. He had other symbols for odd powers and produced tables showing how to multiply these. The words he used for these symbols show Italian and French influence: algebra in the German-speaking world was known as the art of the *coss* (German for *thing*).

By Rudolff's time, the $\pm$ symbols had been in use for around 30 years as a prefix denoting an excess or deficiency in a quantity (profit/loss in accounting); Rudolff began to use

them as *algebraic operations* instead of $\bar{p}$ and $\bar{m}$. A period denoted equals, and he is also credited with the first use of the square-root sign $\sqrt{}$, which is nothing more than a stylized lower-case *r*. This would have been written *in front* of a number, e.g. $\sqrt{23}$ rather than $\sqrt[2]{23}$.

Robert Recorde, England 1557. Introduced the equals sign in *The Whetstone of Witt*, asserting that, ‘No two things are more equal than a pair of parallel lines.’

Francois Viète, France 1540–1603. Before Viète, mathematicians typically described how to solve particular equations algorithmically via examples: e.g. $x^3 + 3x = 14$ rather than the general form $x^3 + bx = c$. Readers were expected to change numbers to fit a required situation while following the same algorithmic structure. Viète pioneered the modern use of abstract constants, using letters to represent both unknowns and constants. As we’ll discuss shortly, this allowed him to investigate how the roots of an equation depend on its constants, a key step in abstract thinking.

Simon Stevin, Holland 1548–1620. *De Thiende* (The Tenth) demonstrated how to calculate using decimals rather than fractions. Stevin arguably completed the journey whereby the concept of *number* subsumed that of *magnitude*. This increased the application of algebra by permitting the numerical description of any geometric magnitude.

William Oughtred, England 1575–1660. Introduced $\times$ for multiplication, though he often simply used juxtaposition. Oughtred combined Viète’s general approach (abstract constants) with symbolic algebra. For instance, he’d write a quadratic equation as $A_q + BA + C = 0$, where A_q means ‘A-squared’ (*A-quadratum* in Latin) and B, C are constants. Its solution, the quadratic formula, would be written

$$A = \sqrt{} : \frac{1}{4}B_q - C : -\frac{1}{2}B$$

In Oughtred’s notation, colons were parentheses.

Thomas Harriot, England 1560–1621. Made several steps towards modern notation including juxtaposition for multiplication and a modern *encompassing* root-sign. For example,

$$\sqrt[4]{cccc + 27aa^3\sqrt{2} + b} \quad \text{meant} \quad \sqrt[4]{c^4 + 27a^2\sqrt[3]{2} + b}$$

René Descartes, France/Holland 1596–1650. Used exponents for powers (a^2, a^3) and solidified the convention of using letters at the end of the alphabet (x, y, z) for unknowns and those at the beginning (a, b, c) for constants.

While modern mathematics uses many specialized symbols ($\emptyset, \Rightarrow, e, \pi$, etc.), basic notation is essentially unchanged from the mid 1600s. This is not to say that all mathematicians uniformly used the most modern notation available: for instance, papers of Leonhard Euler (1700s) used Harriot’s juxtaposition notation for exponents, and some published works from the late 1800s still wrote equations in words.

In this context, it is also worth mentioning Gutenberg's 1439 invention of the printing press. The relative ease of production meant a great increase in the availability of written material. This naturally aided the dissemination of all learning, but led also to the abandonment of some older texts when money could not be found to make an updated printed edition.

Exercises 7.1. *Key concepts: Introduction of Hindu–Arabic numerals to Europe, Notational Development 1200–1650*

1. Repeatedly divide 10 by 5 a total of five times (to $\frac{10}{5^5}$), expressing the results using Fibonacci's notation.
2. What would Nicolas Chuquet have meant by the expression $4\overline{p}R^37\overline{m}R5$?
3. How would William Oughtred have expressed the solution to the quadratic equation $A_q = BA + C$? What about Thomas Harriot?
4. I am owed 3240 *florins*. The debtor pays me 1 *florin* the first day, 2 the second day, 3 the third day, etc. How many days does it take to pay off the debt?
5. (A problem of Antonio de Mazzinghi) Find two numbers such that multiplying one by the other makes 8 and the sum of their squares is 27.
(Hint: let the numbers be $x \pm \sqrt{y}$)

7.2 Polynomials: Cardano, Factorization & the Fundamental Theorem

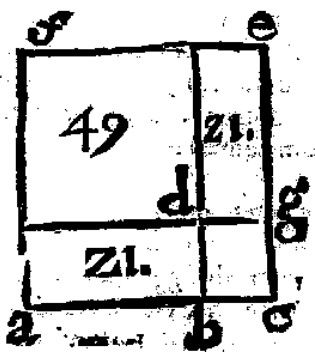
As an example of contemporary algebraic notation and practice, we consider Girolamo Cardano's 1545 *Ars Magna* (*Great Art or the Rules of Algebra*), in which he describes how to solve quadratic and cubic equations.

The first example below (from *Caput V*) is the beginning of Cardano's description of how to solve the equation $x^2 + 6x = 91$ by completing the square: the Latin *quadratum & 6 res aequalie 91* translates as "square and 6 things equals 91." He employs several single-letter abbreviations but still writes in sentences and provides a pictorial justification. In modern algebra, his argument is nothing more than completing the square:

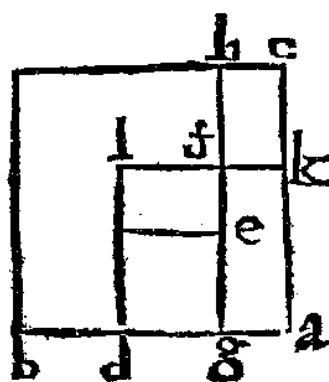
$$\begin{aligned} x^2 + 6x = 91 &\Rightarrow x^2 + 2 \cdot 3x + 3^2 = 91 + 3^2 \\ &\Rightarrow (x + 3)^2 = 100 \\ &\Rightarrow x + 3 = 10 \\ &\Rightarrow x = 7 \end{aligned}$$

where the picture justifies $7^2 + 2 \cdot 21 + 3^2 = 10^2$.

SIT quadratum f d & 6. res (gratiâ exempli) æquale 91. tunc producam d b & d g, quæ sint 3. dimidium 6. numerorum, & complebo quadratum d g b c, indeque productis c g & c b perficiam quadratum a f e c, prout in quarta secundi Elementorum, quia igitur d b ducta in a b ex diffinitione secundi Elementorum producit a d, & ex numero quolibet in rei æstimatione



Completing the square



æquale $\frac{1}{4}$ rei p. 11. duc $\frac{1}{4}$ dimidium numeri rerum in se, fit $\frac{1}{4}$, adde ei 11. fit $11\frac{1}{4}$, accipe 32. quæ est $3\frac{1}{4}$, cui adde $\frac{1}{4}$ dimidium numeri rerum, fit $3\frac{1}{4}$, rei æstimatione. Rursus, fit 1. quadratum æquale 10. rebus p. 6. duc 5. in se dimidium numeri rerum, fit 25. adde ei 6. fit 31. huius 32. adde 5. dimidium numeri rerum, erit rei æstimatione, 31. p. 5. Rursus fit 1. qua-

More quadratic equations

The quadratic algorithms were well-established by this time. On the following page is a picture (second above) reminiscent of al-Khwārizmī (Exercise 6.2.2), after which comes further mathematical notation justifying the solution of another quadratic equation. Even though Rudolff's $\pm$ and $\sqrt{}$ were in use, Cardano wrote almost everything in words augmented with fractional notation and Pacioli's $\bar{p}$ and $\bar{m}$.

As was typical for the time, Cardano describes negative solutions as fictitious; he even writes the square root of -15 at one point, though only to mention its absurdity. He also follows the

Islamic approach of solving a concrete problem of each type rather than proceeding abstractly, though observe the *gratia exempli* ("for the sake of an example") as acknowledgement that the general problem $x^2 + ax = b$ ($a, b > 0$) may be solved identically.

The Cubic Formula

It is for the solution of cubic (and quartic) equations that Cardano is most famous. Below we describe, in modern notation, Cardano's method for solving the cubic equation $x^3 + bx = c$ where $b, c > 0$, though we stress (again) that he gave only *examples* and not a general formula.

Let u, v satisfy $u^3 - v^3 = c$ and $uv = \frac{b}{3}$. Then $x = u - v$ is seen to solve the cubic:

$$\begin{aligned} x^3 + bx &= (u - v)^3 + b(u - v) \\ &= u^3 - 3u^2v + 3uv^2 - v^3 + b(u - v) \\ &= (u^3 - v^3) + (u - v)(b - 3uv) = c \end{aligned}$$

However u and v also satisfy

$$(u^3 + v^3)^2 = (u^3 - v^3)^2 + 4(uv)^3 = c^2 + 4\left(\frac{b}{3}\right)^3$$

We therefore obtain a system of linear equations in the unknowns u^3, v^3 . These are easily solved:

$$\begin{aligned} \begin{cases} u^3 + v^3 = \sqrt{c^2 + 4\left(\frac{b}{3}\right)^3} \\ u^3 - v^3 = c \end{cases} &\Rightarrow \begin{cases} u = \sqrt[3]{\sqrt{\left(\frac{c}{2}\right)^2 + \left(\frac{b}{3}\right)^3} + \frac{c}{2}} \\ v = \sqrt[3]{\sqrt{\left(\frac{c}{2}\right)^2 + \left(\frac{b}{3}\right)^3} - \frac{c}{2}} \end{cases} \\ \Rightarrow x = u - v &= \sqrt[3]{\sqrt{\left(\frac{c}{2}\right)^2 + \left(\frac{b}{3}\right)^3} + \frac{c}{2}} - \sqrt[3]{\sqrt{\left(\frac{c}{2}\right)^2 + \left(\frac{b}{3}\right)^3} - \frac{c}{2}} \end{aligned}$$

Examples Here are three applications of Cardano's method.

1. The equation $x^3 + 6x = 7$ has the obvious solution $x = 1$. Indeed,

$$\begin{aligned} \begin{cases} u^3 - v^3 = 7 \\ uv = \frac{6}{3} = 2 \end{cases} &\Rightarrow (u^3 + v^3)^2 = 7^2 + 4 \cdot 2^3 = 81 \Rightarrow u^3 + v^3 = 9 \\ &\Rightarrow u^3 = \frac{1}{2}(9 + 7) = 8, \quad v^3 = \frac{1}{2}(9 - 7) = 1 \\ &\Rightarrow x = u - v = \sqrt[3]{8} - \sqrt[3]{1} = 2 - 1 = 1 \end{aligned}$$

2. This time we solve $x^3 + 3x = 14$.

$$\begin{aligned} \begin{cases} u^3 - v^3 = 14 \\ uv = \frac{3}{3} = 1 \end{cases} &\implies (u^3 + v^3)^2 = 14^2 + 4 \cdot 1^3 = 200 \implies u^3 + v^3 = 10\sqrt{2} \\ &\implies \begin{cases} u^3 = \frac{1}{2}(10\sqrt{2} + 14) = 5\sqrt{2} + 7 = (\sqrt{2} + 1)^3 \\ v^3 = 5\sqrt{2} - 7 = (\sqrt{2} - 1)^3 \end{cases} \\ &\implies x = u - v = (\sqrt{2} + 1) - (\sqrt{2} - 1) = 2 \end{aligned}$$

As this example shows, Cardano's formula might produce an ugly expression for a simple answer—of course, the cube root of $5\sqrt{2} \pm 7$ is the tip of everyone's tongue!

3. Finally, we consider $x^3 + 3x = 10$. The method works, though in this case the answer is just ugly.

$$\begin{aligned} \begin{cases} u^3 - v^3 = 10 \\ uv = \frac{3}{3} = 1 \end{cases} &\implies (u^3 + v^3)^2 = 10^2 + 4 \cdot 1^3 = 104 \implies u^3 + v^3 = 2\sqrt{26} \\ &\implies u^3 = \sqrt{26} + 5, \quad v^3 = \sqrt{26} - 5 \\ &\implies x = u - v = \sqrt[3]{\sqrt{26} + 5} - \sqrt[3]{\sqrt{26} - 5} \approx 1.6989 \end{aligned}$$

Being unable or unwilling to work directly with negative numbers, Cardano modified his method to solve other cubics such as $x^3 + c = bx$, and moreover described how to remove a quadratic term from a cubic using what is now known as the *Tschirnhaus substitution* ($x = y - \frac{a}{3}$):

$$x^3 + ax^2 + bx + c = \left(y - \frac{a}{3}\right)^3 + a\left(y - \frac{a}{3}\right)^2 + b\left(y - \frac{a}{3}\right) + c = y^3 - \frac{a^2}{3}y + \dots \quad (*)$$

Cardano's student, Lodovico Ferrari, extended the method to solve quartic equations in terms of the solution of a resultant cubic.

Negative Solutions and Complex Numbers

By the late 1500s, mathematicians were mentioning negative solutions to equations. These solutions were usually described as *fictitious*, or *false roots*, but this didn't always prevent them from being investigated. Rafael Bombelli (1526–1572, Rome) even introduced a notation for complex numbers, described their algebra, and showed how they could be used to find solutions to any quadratic or cubic equation. For instance, he wrote $4 + 3i$ and $4 - 3i$ as follows:

4 p di m 3, read 'quattro piu di meno tre' (four plus of minus three), and,
4 m di m 3, 'quattro meno di meno tre.'

Given Bombelli's belief in the fictitiousness of complex numbers, the effort he expended in their honor is extraordinary: this is a prime example of pure abstraction, math for the sake of math!

In modern language, the three roots of Cardano's cubic $x^3 + bx = c$ are

$$u - v, \quad \zeta u - \zeta^2 v, \quad \zeta^2 u - \zeta v$$

where $\zeta = e^{\frac{2\pi i}{3}} = \frac{-1+\sqrt{3}i}{2}$ is a primitive cube root of unity. Together with the Tschirnhaus substitution (*), Cardano's formula therefore solves all cubic equations.

Examples We first return to one of our previous examples, then consider a substitution.

1. Recall that if $x^3 + 3x = 14$, then Cardano's method returns $u = \sqrt{2} + 1$ and $v = \sqrt{2} - 1$. The three solutions to the cubic are therefore

$$u - v = 2$$

$$\zeta u - \zeta^2 v = (\sqrt{2} + 1)\frac{-1 + \sqrt{3}i}{2} - (\sqrt{2} - 1)\frac{-1 - \sqrt{3}i}{2} = -1 + \sqrt{6}i$$

$$\zeta^2 u - \zeta v = (\sqrt{2} + 1)\frac{-1 - \sqrt{3}i}{2} - (\sqrt{2} - 1)\frac{-1 + \sqrt{3}i}{2} = -1 - \sqrt{6}i$$

As a sanity check, it is worth checking these solutions directly.

2. To find a solution to $x^3 + 3x^2 = 3$, we first perform the Tschirnhaus substitution. Define a new variable y via $x = y - \frac{3}{3} = y - 1$, then

$$y^3 - 3y^2 + 3y - 1 + 3(y^2 - 2y + 1) = 3 \implies y^3 - 3y = 1 \quad (b = -3, c = 1)$$

which is the standard cubic with $b = -3, c = 1$. Cardano would have applied his modified algorithm for $y^3 = 3y + 1$, but we'll just use negative numbers in the original:

$$\begin{aligned} \begin{cases} u^3 - v^3 = 1 \\ uv = -\frac{3}{3} = -1 \end{cases} &\implies (u^3 + v^3)^2 = 1^2 + 4(-1)^3 = -3 \\ &\implies u^3 + v^3 = \sqrt{3}i \\ &\implies u^3 = \frac{1}{2}(\sqrt{3}i + 1), \quad v^3 = \frac{1}{2}(\sqrt{3}i - 1) \\ &\implies x = y - 1 = u - v - 1 = \sqrt[3]{\frac{\sqrt{3}i + 1}{2}} - \sqrt[3]{\frac{\sqrt{3}i - 1}{2}} - 1 \end{aligned}$$

This is ugly! If you know Euler's formula and you choose a compatible pair of cube roots (remember that $uv = -1$), this will evaluate to one of three real numbers: the positive solution is in fact $x = 2 \cos 20^\circ - 1 \approx 0.8794$.

Factorization & the Fundamental Theorem of Algebra

By the late 1500s, Viète's abstraction allowed him to streamline Cardano's methods, as well as to investigate the relationship between the coefficients of a polynomial and its roots. For Viète the roots had to be positive, but later improvements by Thomas Harriot and Albert Girard (1629) applied this to all polynomials with any roots.

Here is the approach for the quadratic $ax^2 + bx + c = 0$, with roots r_1, r_2 :

$$\begin{aligned}(r_1 - r_2)(a(r_1 + r_2) + b) &= a(r_1^2 - r_2^2) + b(r_1 - r_2) \\ &= (ar_1^2 + br_1 + c) - (ar_2^2 + br_2 + c) = 0\end{aligned}$$

Provided the roots are distinct,⁵⁰ we conclude that

$$\frac{b}{a} = -(r_1 + r_2), \quad \text{and} \quad \frac{c}{a} = -r_1^2 - \frac{b}{a}r_1 = -r_1^2 + (r_1 + r_2)r_1 = r_1r_2$$

These are the quadratic version of what are now known as *Viète's formulas*; versions exist for polynomials of every degree. Their use amounts to an early form of factorization.

Example By inspection, we spot that $3x^2 - 2x - 1 = 0$ has $r_1 = 1$ as a root. Applying Viète's formulas,

$$r_1 + r_2 = -\frac{b}{a} = \frac{2}{3} \implies r_2 = -\frac{1}{3} \quad \text{or alternatively} \quad r_1r_2 = \frac{c}{a} = -\frac{1}{3} \implies r_2 = -\frac{1}{3}$$

Think about the relationship between this approach and factorization!

A nice side-effect is a method for obtaining the quadratic formula by solving a pair of simultaneous equations analogous to Cardano's cubic approach:

$$\begin{cases} r_1 + r_2 = -\frac{b}{a} \\ r_1 - r_2 = \sqrt{(r_1 + r_2)^2 - 4r_1r_2} = \sqrt{\left(\frac{b}{a}\right)^2 - 4\frac{c}{a}} \end{cases} \implies r_1, r_2 = -\frac{b}{2a} \pm \frac{1}{2}\sqrt{\left(\frac{b}{a}\right)^2 - 4\frac{c}{a}}$$

Viète's formulas were central to the later development of Galois Theory (1830) and the Abel–Ruffini theorem regarding the insolubility of quintic and higher-degree polynomials.

The remaining key results regarding solutions of polynomials also appeared around this time:

Fundamental Theorem of Algebra Including multiplicity, a degree n polynomial has n complex roots. Girard offered the first version in 1629, though a complete proof didn't appear until the work of Argand, Cauchy and Gauss in the early 1800s.

Factor Theorem In 1637, Descartes proved a familiar result, essentially via long-division:

$$p(r) = 0 \iff p(x) \text{ is divisible by } x - r.$$

⁵⁰In fact the formulas hold even when $r_1 = r_2$.

For instance, here is Descartes' method to find the roots of $p(x) = x^3 + 2x^2 - 13x + 10$:

- Observe that $p(1) = 0$. Thus $x - 1$ is a factor.
- Use long-division to obtain $p(x) = (x - 1)(x^2 + 3x - 10)$.
- $q(x) = x^2 + 3x - 10$ has $q(2) = 0$; divide to get $q(x) = (x - 2)(x + 5)$.
- The roots of $p(x)$ are therefore 1, 2 and -5 (Descartes called this last a *false root*).

We'll return to Descartes in the context of Analytic Geometry in the next chapter.

Exercises 7.2. *Key concepts: Cardano's method for $x^3 + bx = c$, Abstraction via arbitrary constants, Investigating the relationship between coefficients and roots*

- (a) Find Viète's formulas for the polynomial $p(x) = x^3 + ax^2 + bx + c$ with roots r_1, r_2, r_3 ; that is, find the coefficients a, b, c in terms of the roots.
(Hint: Multiply out $p(x) = (x - r_1)(x - r_2)(x - r_3)\dots$)
- (b) Solve $x^3 - 6x^2 + 9x - 4 = 0$ using Girard's method: first, determine one solution by inspection, then use Viète's formulas for the cubic to investigate the relationship between the remaining roots.
- (a) Apply Cardano's method to the equation $x^3 + 6x = 20$.
(Hint: to finish, compute $(1 + \sqrt{3})^3$)
- (b) If $b, c > 0$, Cardano's method finds a single positive solution to $x^3 + bx = c$. Explain why such an equation always has exactly one real solution which is moreover positive.
- Prove that if t is a root of $x^3 = cx + d$, then

$$r_1 = \frac{t}{2} + \sqrt{c - \frac{3t^2}{4}} \quad \text{and} \quad r_2 = \frac{t}{2} - \sqrt{c - \frac{3t^2}{4}}$$

are both roots of $x^3 + d = cx$. Use this to solve $x^3 + 3 = 8x$.

- Consider the cubic equation $aaa - 3raa + ppa = 2xxx$ (as written by Harriot). Show that the substitution $a = e + r$ reduces this to an equation without a square term.
As an example, reduce the equation $aaa - 18aa + 87a = 110$ to a cubic in e without a square term. Find all three solutions (in e) and thus the solutions to the original equation (in a).
- Find all roots of the cubic $p(x) = 2x^3 - 3x^2 - 3x + 2$ using Descartes' factor theorem.
- Use Cardano's method (with Tschirnhaus substitution and complex numbers!) to find the solutions to the equation $x^3 + 4 = 3x^2$. Verify that you get the same solutions using Descartes' factor theorem.
(Hint: all solutions are integers!)
- (If you are very comfortable with complex numbers) Use Euler's formula to verify that the equation $x^3 + 3x^2 = 3$ has the positive solution $x = 2 \cos 20^\circ - 1$. It also has two negative real solutions: find them!

7.3 Astronomy, Trigonometry & Kinematics in the Renaissance

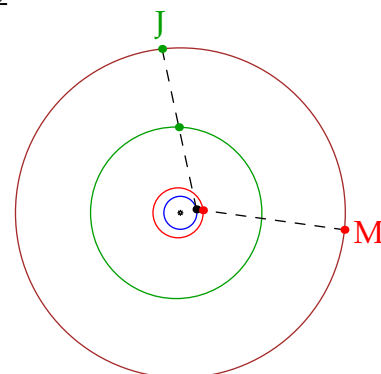
Early European Trigonometry to Copernicus

Early trigonometry came to Europe via translations of Ptolemy's *Almagest* and its Islamic improvements. In 1463, Johannes Müller (a.k.a. Regiomontanus,⁵¹ 1436–76) provided an axiomatic update to this canon in *De Triangulis Omnimodius* (*Of all kinds of triangles*). Though the title mentions triangles, his approach was that of the ancients: circle-based, using chords and half-chords. Regiomontanus was a renowned astronomer; his tracking of a comet from late 1471 to spring 1472 provided evidence, at the time controversial, that objects could move between the “fixed” heavenly spheres of ancient Greek theory.

Domenico Novara (1454–1504), Regiomontanus' student, inherited much of his unpublished work. He also studied with Luca Pacioli in Florence and became an astronomer at the University of Bologna. He is best known, however, for advising a young Pole, Nicolaus Copernicus (1473–1543), who studied in Bologna from 1496.

Copernicus' ostensible purpose was to join the priesthood, but his interests drew him to mathematics and astronomy. His investigations forced him to conclude that Ptolemy's geocentric model could not be reconciled with astronomical observation. His magnum opus, *De revolutionibus orbium celestium* (*On the revolutions of the heavenly spheres*), was published a year after his death. Within, Copernicus describes how one would calculate (chords and angles) within a *heliocentric* (*sun-centered*) model. This approach, Copernicus believed, was the obvious solution to the problem of *retrograde motion* that had plagued the ancient Greeks.⁵²

The animation demonstrates the heliocentric model as upgraded later by Kepler (Copernicus proposed circular rather than elliptic motion). Viewed from **Earth**, the retrograde motion of **Mars** and **Jupiter** are easily explained. The **outer circle** represents the ‘fixed stars’ against which the motion of the planets are observed. Each step corresponds to three days of motion. Retrograde motion is observed when Earth, being closer to the sun, catches and overtakes Mars or Jupiter. This is most obvious with Mars since the orbits are so similar in size.



Copernicus' work on heliocentrism is often described as a revolution, though it was not perceived so at the time. *De revolutionibus* was dedicated to the Pope, welcomed by the Church, and used by Vatican astronomers to aid in calculation. It was a difficult text which resulted in a narrow readership; his work was therefore unthreatening to contemporary dogma. Moreover, Copernicus did not present heliocentrism as reality nor did he advocate for overturning long-held beliefs; in his presentation, heliocentrism is merely a simple model that seems to produce good predictions. The fact, however, that he did not publish during his lifetime suggests that he knew that he feared a revolution was coming. Within a century, the Copernican theory had found its bulldog in Galileo, and conflict between Science and the Church became unavoidable.

⁵¹His grand-sounding name is a latinization of his birthplace: Königsberg (*King's Mountain*), Bavaria, Germany.

⁵²Copernicus wasn't the first to suggest a heliocentric model. Several ancient Greek scholars embraced the idea; Aristarchus of Samos (c. 310–230 BC) is credited with its first serious presentation. Aristarchus' views, however, were rejected by mainstream Greek scholars, and it is likely Copernicus never encountered his work.

Trigonometry is finally about triangles!

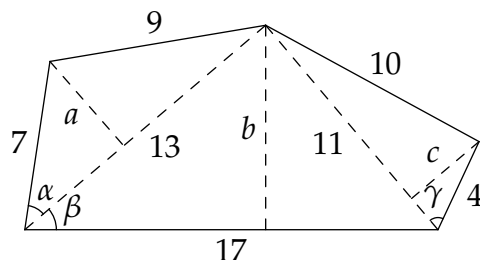
Georg Rheticus (1514–1574) was a student of Copernicus, who helped posthumously to publish the latter's work. In the late 1500s, Rheticus started defining trigonometric functions purely in terms of triangles, referring to the *perpendicularum* (sine) and *basis* (cosine) of a right-triangle with fixed hypotenuse.

The modern term finally arrives in 1595. In *Trigonometriæ*, Bartholomew Pitiscus purposefully sets out to solve problems related to triangles. Pictured is the title page from part of the second edition (1600—MDC in Roman numerals): note also the subtitle, which translates as “The dimensions of triangles.” Both Rheticus and Pitiscus had problems which look very familiar to modern readers, such as solving for unknown sides of triangles.



Example (Pitiscus) A field has five straight edges of lengths 7, 9, 10, 4 and 17 in order. The distance between the first and third vertices is 13, and between the third and fifth 11. What is the area of the field?

The problem can easily be solved using Heron's formula, but Pitiscus opts for trigonometry. Here is a modernized version that depends on applying the law of cosines to the three large triangles.



$$\cos \alpha = \frac{7^2 + 13^2 - 9^2}{2 \cdot 7 \cdot 13} = \frac{137}{182}$$

$$\cos \beta = \frac{17^2 + 13^2 - 11^2}{2 \cdot 17 \cdot 13} = \frac{337}{442} \quad \cos \gamma = \frac{4^2 + 11^2 - 10^2}{2 \cdot 4 \cdot 11} = \frac{37}{88}$$

The values of α, β, γ and therefore the altitudes a, b, c of the three major triangles could be read off a table, or computed exactly using Pythagoras':

$$a = 7 \sin \alpha = 7 \sqrt{1 - \cos^2 \alpha} = \frac{7}{182} \sqrt{182^2 - 137^2} = \frac{3}{26} \sqrt{1595}$$

$$b = 13 \sin \beta = \frac{1}{34} \sqrt{81795} \quad c = 4 \sin \gamma = \frac{5}{22} \sqrt{255}$$

The total area is therefore

$$A = \frac{1}{2} (13a + 17b + 11c) = \frac{1}{4} (3\sqrt{1595} + \sqrt{81795} + 5\sqrt{255}) \approx 121.4$$

Kepler's Laws of Planetary Motion

Johannes Kepler (1571–1630) was a student of Tycho Brahe⁵³ for the last two years of Brahe's life. He inherited Brahe's position, decades-worth of astronomical data, and his philosophy on the importance of observation-based theory.

Kepler also embraced the mystical Pythagorean view that nature reflects harmony, a belief that partly drove his scientific pursuits. To Kepler, any observation of a natural, simple ratio was something of great import. For example, in observing that the daily movement of Saturn at its furthest point from the sun was roughly 4/5 of that at its nearest point, his temptation was to assume that 'roughly' must be 'exactly.'

Thanks to Brahe, Kepler had good data for roughly thirteen orbits of Mars and two of Jupiter. From these data, he posited three laws. These laws are empirical observations rather than the result of mathematical proof. However, the process of their discovery demonstrates Kepler's tremendous mathematical abilities.

The First Law *Planets move in ellipses with the sun at one focus.*

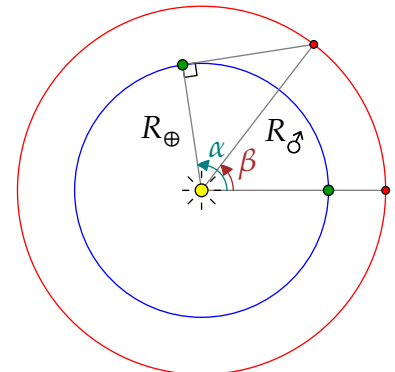
Kepler started by assuming essential correctness of the Copernican model in that all planets exhibit uniform circular motion round the sun. Due to the relative wealth of data, he devoted most effort to Mars. Kepler wanted to compute both its *orbital radius* $R_{\text{♂}}$ and *orbital period* $T_{\text{♂}}$ (Earth years). The essential calculation relies on two astronomical phenomena.

Opposition **Earth** lies directly between the sun and **Mars**. Equivalently, Mars is directly overhead at midnight.

From observations, Kepler knew the time between successive oppositions to be approximately $S = 2.14$ years (780 days).

Quadrature The Earth-Mars-Sun triangle is right-angled at Earth. Mars is roughly on the horizon at midnight.

The time between opposition and quadrature is approximately 100 days.



Since Earth orbits faster than Mars, it must have completed *exactly one more orbit* over the $S = 2.14$ years between oppositions, during which time Mars completes $\frac{S}{T_{\text{♂}}}$ orbits. Kepler concludes,

$$S = \frac{S}{T_{\text{♂}}} + 1 \implies T_{\text{♂}} = \frac{S}{S - 1} = \frac{2.14}{1.14} = 1.88 \text{ years}$$

In the 100 days between quadrature and opposition, Earth and Mars travel through angles

$$\alpha \approx \frac{100 \times 360}{365} = 98.6^\circ, \quad \beta = \frac{\alpha}{T_{\text{♂}}} \approx 52.5^\circ$$

⁵³Tycho Brahe (1546–1601) was a Danish astronomer who worked for Austro-Hungarian Emperor Rudolph II in Prague for 25 years, producing a wealth of accurate astronomical measurements. While these helped burnish the Copernican theory, he is also known for his 1572 observation of a *nova* (a *new* star that later disappeared, now understood to be the death of a distant star), and then a comet in 1577; both provided more evidence for the changeability of the heavens.

The orbital radius of Mars is therefore

$$R_{\text{♂}} = \sec(\alpha - \beta)R_{\oplus} \approx 1.44R_{\oplus}$$

Earth's orbital radius $R_{\oplus}$ (one *astronomical unit* [AU]), was not known accurately in Kepler's time, so he only knew the *relative* distance to Mars.⁵⁴

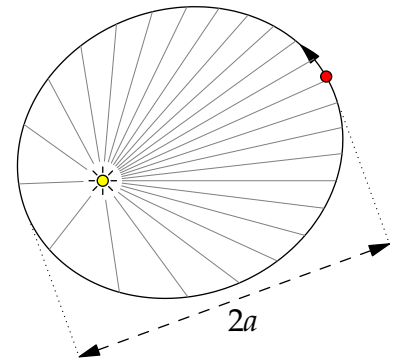
There was enough variation in the data for Kepler to realize that purely circular orbits were not quite right: times between oppositions, and between opposition and quadrature are not constant; other observations didn't quite fit with the predicted location of Mars. Initially he altered his model to reflect Earth's slightly non-circular orbit. Mimicking the ancient Greeks, he first tried an equant model, offsetting the center of the orbit slightly from the sun. Despite this, he failed to fit his data for Mars to pure circular motion.

Abandoning circles, Kepler now permitted planets to move in ovals. He decided to approximate Mars' orbit with an ellipse and set out to calculate its parameters, stumbling on an almost perfect match when the sun was placed at a focus. The geometric significance of the focus was exactly the natural beauty sought by Kepler. Having now established the first law for Mars, he repeated the exercise for the other known planets (Mercury, Venus, Jupiter, Saturn) as well as he could given his inferior data.

The Second Law *The orbital radius sweeps out equal areas in equal times.*

The pictured sectors are swept out in equal times and have the same area. In modern language, this is the conservation of angular momentum.

To fit his elliptical orbits, Kepler noticed that planetary velocity seemed inversely proportional to the distance from the sun ($v = \frac{k}{r}$). He used this to approximate the area of a sector swept out by the radius vector: over an infinitesimal time-interval Δt , a planet travels a distance $v\Delta t$, sweeping out an infinitesimal triangle of area $\frac{1}{2}rv\Delta t = \frac{1}{2}k\Delta t$.



Kepler had no justification beyond the fact that it seemed to fit the data. In particular, he did not know why a planet should move more slowly when further from the sun.

The Third Law *The square of the period is proportional to the cube of the semi-major axis: $T^2 \propto a^3$.*

This was stated with very limited analysis. Given the relative sizes of each orbit and its period, this follows from little more than inspired guesswork.

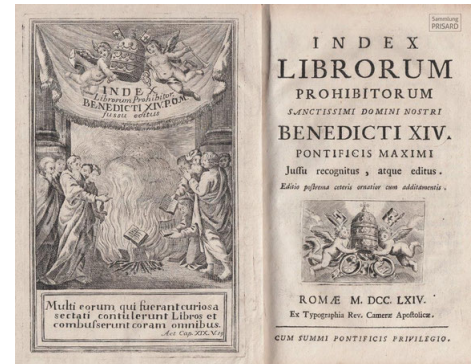
Kepler's discoveries were revealed over many years in several texts, and his magnum opus *Epitome astronomiæ Copernicanæ* was published in 1621. Within a century, Issac Newton provided a mathematical justification of Kepler's laws based on calculus and his own axioms: an inverse square law for gravitational acceleration and his own three laws of motion.

⁵⁴Following Ptolemy, Brahe thought $R_{\oplus}$ to be around a tenth of its true value, though Kepler thought it to be at least three times larger. In 1659, Christiaan Huygens found $R_{\oplus}$ to within 3 percent.

A Religious Interlude: Protestantism, the Counter-Reformation & Calendar Reform

By the mid 1500s, a new brand of Christianity was on the rise, Protestantism, its spread aided by the new technology of mass-printing (1440s onward). The Catholic Church's response is known as the *counter-reformation*.⁵⁵

Of particular interest to science was the *Index Librorum Prohibitorum* (from c. 1560), a list of books contradicting Church doctrine. Texts advocating heliocentrism were banned, since it challenged both longstanding doctrine and specific Biblical passages ("The sun stopped in the middle of the sky...", Joshua 10:13). Kepler's *Empitome* was prohibited upon publication, though his location far from Rome ensured his relative safety. Indeed, the ultimate effect of this crackdown was the ceding of scientific power to northern (Protestant) Europe where papal diktat had less effect.



Front page of the *Index*, 1764

The Gregorian Calendar In contrast to the anti-scientific book-banning fervor of the counter-reformation, Pope Gregory XIII (reign 1572-85) is famous for shepherding an astounding scientific achievement: calendar reform.

By 1500, astronomers knew the solar year to be roughly $11\frac{1}{4}$ minutes shorter than the $365\frac{1}{4}$ days of the Julian calendar: named for Julius Caesar, the Julian year has 365 days with a leap-day added every four years. Since the First Council of Nicea (325), Easter had been defined as the Sunday after the first full moon after the vernal equinox (March 21st). By 1500, however, the equinox was observed to be happening around March 10th, raising concern that Easter was being celebrated on the wrong day. Calendar reform and improved astronomical modeling thus became an important Church project.

A century of effort and debate⁵⁶ resulted in the modern *Gregorian calendar*, which Gregory imposed in all Catholic countries in 1582. This made two major changes to the Julian calendar:

- Centuries are leap-years only when divisible by 400 (e.g., 1900 was not a leap year).
- To correct the ten-day deficit, Gregory eliminated October 5th–14th, 1582.

The Gregorian calendar is astonishingly accurate, losing only one day every 3000 years. Since it emanated from Rome, Protestant parts of Europe took decades or centuries to adopt the new system. The Eastern Orthodox Church still computes Easter using the Julian calendar, which is now 13 days behind the Gregorian.⁵⁷

⁵⁵Martin Luther's *Ninety-five Theses* (1517) is generally considered the start of the Protestant Reformation. Over the next 150 years, Europe saw gruesome religious wars and millions of deaths as various countries broke from Rome. With its emphasis on a personal relationship with God unmediated by central authority, Protestantism has been instrumental in making European and North-American culture more individualistic.

⁵⁶Pope Sixtus IV tried to recruit Regiomontanus to the cause in 1475, though the mathematician died before being able to contribute. Copernicus was among those invited to consider proposals in the early 1500s, though he distanced himself, perhaps because he knew that his developing heliocentric ideas would not be accepted. The final design was largely worked out by Aloysius Lilius and Christopher Clavius.

⁵⁷Because the computation of Easter is downstream from the calendrical equinox (March 21st), Orthodox Easter occurs between one and five weeks after Gregorian Easter.

Galileo Galilei (1564–1642)

Based in northern Italy, Galileo was close to the center of Church power. Unlike Copernicus and Kepler, he openly challenged its orthodoxy. While undoubtedly a great mathematician, he is more importantly considered the father of the scientific revolution due to his reliance on experiment and observation. He famously observed Jupiter's moons with a telescope of his own invention, noting that objects orbiting an alien body ran counter to Ptolemaic theory. Skeptics, when shown these images, preferred to assert that it must be somewhere *inside* the telescope!

In 1632 Galileo published *Dialogue Concerning the Two Chief World Systems*, a Socratic discussion between three characters: Salviati argued for Copernicus, Simplicio was for Ptolemy, and Sagredo was an independent questioner. The character of Simplicio was provocatively modeled on conservative philosophers who refused to consider experiments; even worse, he bore a notable resemblance to the Pope. Salviati almost always came out on top and Simplicio was made to appear foolish. The text contributed to Galileo's conviction for heresy; all his publications, past and future, were banned, and he spent the remainder of his life under house arrest.

Despite these restrictions, Galileo's discoveries continued to be distributed by his supporters. His most important scientific text, *Discourses Concerning Two New Sciences* (materials science and kinematics) was smuggled out of Italy to be published in Holland in 1638. In this book, he resurrects his characters from *Two Chief World Systems* and famously refutes Aristotle's claim that heavier objects fall more rapidly than lighter ones.⁵⁸ Here are two results from this text.

Theorem. *If acceleration is uniform, then the average speed is the average of the initial and terminal speeds.*

Proof. Galileo argues pictorially, as shown on the right.

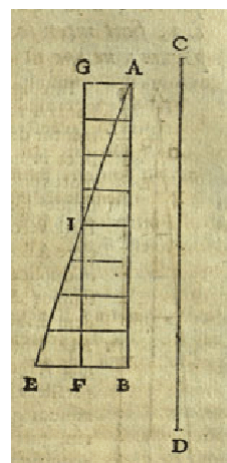
In essence, $\overline{CD}$ is the time-axis, increasing downward. Velocity is measured horizontally from $\overline{CD}$ to $\overline{AE}$, of which I is the midpoint.

Galileo argues that the distance travelled is the area between $\overline{AE}$ and the time-axis. Since $\triangle ABE$ and $\square ABFG$ have the same area, the distance travelled plainly equals the area of the rectangle between $\overline{GF}$ and the time-axis. Since $|AG| = \frac{1}{2}BE$, the velocity corresponding to $\overline{GF}$ is the average of those corresponding to A and E .

Here is the calculation using modern algebra. Suppose the object has velocities v_A and v_B as it passes A and B respectively. Also let $t = |AB|$. Then the distance travelled is

$$\begin{aligned} v_{av}t &= v_A t + \text{area}(\triangle ABE) = v_A t + \text{area}(\square ABFG) \\ &= v_A t + \frac{1}{2}(v_B - v_A)t = \frac{1}{2}(v_A + v_B)t \end{aligned}$$

whence v_{av} is the average of the initial and final speeds. ■



⁵⁸Supposedly by dropping weights off the leaning tower of Pisa, though take this story with a pinch of salt...

Corollary. *A falling object dropped from rest traverses distance in proportion to time-squared,*

$$d_1 : d_2 = t_1^2 : t_2^2$$

This is Galileo's version of the well-known kinematics formula $d = \frac{1}{2}gt^2$.

Proof. Let d_1, d_2, v_1, v_2 represent the distances travelled and the speeds of the dropped body at times t_1 and t_2 . Since acceleration is uniform,

$$v_1 : v_2 = t_1 : t_2$$

By Galileo's Theorem,

$$d_1 = \frac{0 + v_1}{2}t_1 = \frac{1}{2}v_1t_1, \quad \text{and} \quad d_2 = \frac{1}{2}v_2t_2$$

Putting everything together, we obtain

$$d_1 : d_2 = v_1t_1 : v_2t_2 = t_1^2 : t_2^2$$

as required. ■

Galileo follows this result by decomposing the motion of a projectile into horizontal (uniform speed) and vertical (uniform acceleration) components, thereby proving that projectiles follow parabolic paths.

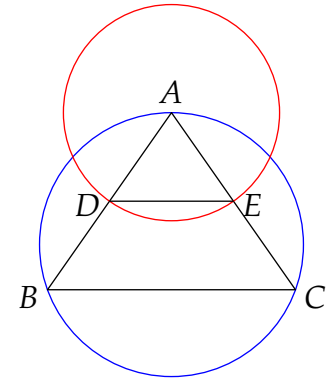
Galileo covered several other important mathematical topics, including, as we'll see later, early ideas from calculus which also drew the condemnation of the Church. While his mathematical ideas were cutting-edge for the time, it is his insistence on testing theory against data that makes him a true scientific revolutionary. By 1600, very few of Aristotle's easy-to-refute claims had been rejected due to experimental testing—the hostility provoked by Galileo helps explain why. This is the core of the scientific revolution: primacy is given to experiment and observation over 'ancient wisdom,' regardless of its source.

The Catholic Church's stance with regard to heliocentrism has slowly modernized. Edited versions of Galileo's works were permitted from 1718, and general texts advocating heliocentrism from 1758. The works of Kepler and Copernicus were removed from *Index* in 1835, with the entire project being discontinued in 1966. Galileo was finally cleared of heresy in 1992.

Exercises 7.3. Key concepts: Copernicus & Heliocentrism, Trigonometry about measuring triangles, Kepler's Laws, Galileo (experimentation, observation and kinematics)

1. Compute today's date in the Julian calendar and explain your calculation.
2. (A problem of Copernicus) Given the three sides of an isosceles triangle, to find the angles.

Suppose $\overline{AB}$ and $\overline{AC}$ are the equal legs of the triangle. Circumscribe a **circle** around the triangle and draw **another** with center A and radius $\overline{AD} = \frac{1}{2}\overline{AB}$.



- (a) Why does Copernicus introduce the second circle?
 - (b) Explain why the ratio of each of the equal sides to the base of $\triangle ABC$ equals that of the radius $\overline{AD}$ to the chord $\overline{DE}$.
 - (c) If $|AB| = |AC| = 10$ and $|BC| = 6$, use modern trigonometry to find the three angles of the original triangle.
3. Verify that Heron's formula gives the same solution to Pitiscus' problem on page 96.
 4. Given that Earth's orbital period is 1 year and that the mean distance of Mars from the sun is 1.524 times that of Earth, use Kepler's third law to determine the orbital period of Mars. (As we've seen, Kepler determined T_{J} before discovering his third law)
 5. According to Kepler's second law, at what point in a planet's orbit will it be moving fastest?
 6. Galileo states the following.

A projectile fired at an angle $\alpha = 45^\circ$ above the horizontal at a given initial speed reaches a distance of 20,000. Then, with the *same* initial speed it will reach a distance of 17,318 when $\alpha = 60^\circ$, or $\alpha = 30^\circ$.

Verify this statement. If you want a challenge, try to do without the standard Physics formulæ and instead use ratios!